Write and Represent Inequalities

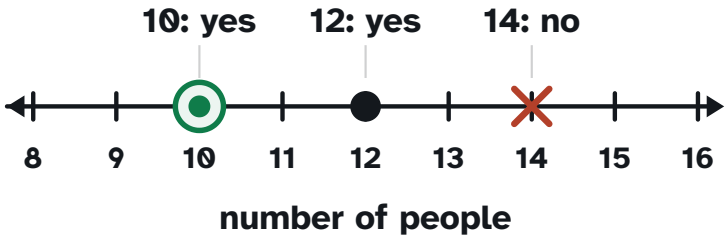
Show an inequality on a number line with an open or closed circle.

Name _____ **Date** _____

Tonight's spark

An elevator sign says: "No more than 12 people." You draw the allowed counts on a number line. Is the circle at 12 filled in, or left empty?

Guess first: can exactly 12 people ride, yes or no?



Our class model: the closed-or-open circle test

Try it

Work it out on paper. Write one answer on each line.

1 **On a number line, what does a closed (filled) circle at 4 tell you?**

A. 4 is not a solution **B.** 4 is a solution **C.** Only 4 is a solution
D. 4 is the smallest number

My answer _____

2 **Which inequality needs an open circle at 6?**

A. $x \geq 6$ **B.** $x \leq 6$ **C.** $x > 6$ **D.** $x = 6$

My answer _____

3 **Which inequality needs a closed circle at 9?**

A. $x \leq 9$ **B.** $x < 9$ **C.** $x > 9$

My answer _____

4 **The graph of x > 2 has an open circle at 2. Which way is it shaded?**

A. Left, toward smaller numbers **B.** Both directions **C.** Nowhere, it is only a dot
D. Right, toward larger numbers

My answer _____

5 **A graph has a closed circle at 5 and shading to the left. Which inequality does it show?**

A. $x < 5$ **B.** $x \leq 5$ **C.** $x > 5$ **D.** $x \geq 5$

My answer _____

Talk about it

Does the number at the circle count? How does the circle show that?

Home mission

Use a ruler as a number line. Say: "No more than 12 inches." Put a coin on 12. Should the coin count as a closed or open circle? Which way do you slide your finger?

If we get stuck, ask

- Does the boundary number itself count as a solution?
- Which words tell you that — "at most", "more than"?
- Which direction do the allowed values go?

Do it on a screen
Scan or type the link. Same lesson, with hints.
eduwonderlab.com/lessons/8-4-part2/homework.html

How did it feel? (optional)
Tricky Getting there Got it